

MATH 195: 9/29

EXPONENTIAL GROWTH AND DECAY

Exponential functions have the property that their *rate of change* or *slope* is proportional to their value. For any base $a \neq 1$, there is a constant K_a so that the slope of a^x at x is $K_a a^x$.¹ There is a number e so that $K_e = 1$. This number is special and important because e^x (and its vertical scalings/reflections) is the only function whose slope is itself. This makes it useful in calculus and applications of calculus.

$$e \approx 2.71828182845904523536 \dots$$

is *irrational*, meaning it cannot be written as a fraction of whole numbers. Moreover it is *transcendental*, meaning that it's not a zero of a polynomial with whole number coefficients. To precisely define e requires ideas from calculus. Here is one way, which requires the notion of a *limit* to make precise:

$$\text{as } n \rightarrow \infty, \quad \left(1 + \frac{1}{n}\right)^n \rightarrow e.$$

Any quantity whose rate of change is proportional to its size can be expressed by a function as

$$P(t) = P_0 e^{rt},$$

where P_0 and r are constants. As we will see, P_0 represents the *initial value* of the quantity and r says something about the rate of change.

A bacteria culture begins with 1000 cells. After one day, there are 1350 cells. Assume the growth rate is proportional to the number of cells, so you can model it as

$$B(t) = B_0 e^{rt} \text{ bacteria cells after } t \text{ days.}$$

- (1) Use the initial value information to solve for B_0 . [Hint: plug in the value at $t = 0$ and solve for B_0 .]
- (2) Use desmos or another graphing calculator to plot $B(t)$, based on the not-yet known constant r . In desmos, you can type `B_0 e^rx` and it will give you the option to create a slider for r to change its value.
- (3) Using the graphing calculator as an aid, estimate the value for r that makes $B(1) = 1350$.
- (4) Now that you know r , use your function to estimate how many cells will be in the culture after 5 days. After 10 days.
- (5) Using the graphing calculator, estimate how long it will take for the culture to reach 50 thousand cells.
- (6) How would it go if you tried to algebraically solve for when the culture reaches 50 thousand cells? Try to solve $B(t) = 50000$ for t . When you get to a spot where you need something we haven't talked about yet in this class, what tool do you need to proceed? Explain.

Plutonium-239 is a radioactive isotope of plutonium. It undergoes radioactive decay, with a half-life of about 24 thousand years. That is, after 24 thousand years half of the atoms will have decayed to a different element. A quantity of plutonium-239 over time can be modeled as

$$P(t) = P_0 e^{rt},$$

where r is negative.

- (1) Suppose you start with 1 kilogram of plutonium-239. Use this to determine the value of P_0 .
- (2) Use desmos or another graphing calculator to plot $P(t)$, based on the not-yet known constant r .
- (3) Using the graphing calculator as an aid, estimate the value for r that makes $B(24000) = 0.5$.
- (4) Using the graphing calculator as an aid, estimate how long it will take for 80% of your plutonium-239 to decay.
- (5) How would it go if you tried to algebraically solve for r ? Try to solve $B(24000) = 0.5$ for r . When you get to a spot where you need something we haven't talked about yet in this class, what tool do you need to proceed? Explain.

¹If you've seen logarithms before, $K_a = \ln(a)$. When you take calculus you will see where this comes from.

There are limitations to the use of exponential functions to model population growth, as you now explore.

In 2000, the global population was estimated to be approximately 6.15 billion people. In 2010, that increased to approximate 6.99 billion. Let's model this with an exponential function

$P(t) = P_0 e^{rt}$ billion people t years after 2000.

- (1) Similar to before, find P_0 exactly and use a graphing calculator to estimate r .
- (2) Use your function $P(t)$ to predict the global population in 2100. In 2200. In 2300. In 2400.
- (3) Explain why your predictions are bad.

There are other models for population growth which give better long-term predictions, but they come at the cost of a more complicated formula.

One such model is *logistic growth*:

$$P(t) = \frac{L}{1 + \frac{L - P_0}{P_0} e^{-rt}},$$

where P_0 is the initial population, r determines the rate of change, and L is the *carrying capacity*, the largest population that can be supported.

An environment can support a rabbit population of 2000. The initial population of rabbits is 100. Suppose after 10 months the population has increased to 140 rabbits.

- (1) Knowing the initial population is 100 and the carrying capacity is 2000, use a graphing calculator to estimate the value for r that gives $P(10) = 140$. Like before you can put in a slider for r to make it easier to test different values.
- (2) Once you know r , using your function and its graph to estimate the following:
 - (a) The population after 50 months.
 - (b) How long it will take for the population to reach 1000 rabbits.
 - (c) What value $P(t)$ approaches as $t \rightarrow \infty$.
- (3) Logistic models still have limitations and inaccuracy. What do you think are some reasons why this model might give inaccurate predictions?